

LongSAL: A Longitudinal Search as Learning Study with University Students

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ABSTRACT

Learning, or addressing a gap in one's knowledge, is an important motivator behind information-seeking activities. The Search as Learning research community advocates for transforming information search systems into educational platforms to support learning and sensemaking. Current search systems have yet to adapt to this function. One crucial step is identifying behavioural patterns that distinguish individuals who gain more knowledge from those who gain less. Previous research has mainly focused on short-term, laboratory-based studies of human-information interaction, which researchers believe is insufficient for understanding the prolonged process of learning. In response, we conducted LongSAL, an exploratory longitudinal study examining the long-term information searching behaviour of university students over a semester. Our preliminary results indicate that participants with higher levels of metacognition and self-regulation exhibit greater efficiency in their searching as learning process. Specifically, they visit fewer webpages, and spend less time dwelling on them, yet receive similar grades compared to participants who spend more time dwelling, and visiting more webpages. These findings will help develop improved search systems that foster learning and sensemaking.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **User models**; • **Information systems** → **Users and interactive retrieval**.

KEYWORDS

search as learning; longsal; longitudinal study; human information behaviour; interactive IR experiments; remote IR studies; hii; iir;

ACM Reference Format:

Nilavra Bhattacharya. 2023. LongSAL: A Longitudinal Search as Learning Study with University Students. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems (CHI EA '23)*, April 23–28, 2023, Hamburg, Germany. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3544549.3583948>

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CHI EA '23, April 23–28, 2023, Hamburg, Germany

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ACM ISBN 978-1-4503-9422-2/23/04.

<https://doi.org/10.1145/3544549.3583948>

1 INTRODUCTION AND BACKGROUND

“Learning today is ... about navigation, discernment, induction, and synthesis” of the wide body of information on the Internet present ubiquitously at every student's fingertips [10]. Learning, or addressing a gap in one's knowledge, has been well established as an important motivator behind information-seeking activities [15, 28, 36, 41]. As a result, search engines play a crucial role in modern education. However, it is critical to remember the many limitations of today's search engines: “they do not help us know what we want to know; they do not help us choose the right words to find it; they do not help us know if what we've found is relevant or true; and they do not help us make sense of [the retrieved information]” [21]. The Searching as Learning research community advocates that information search systems should be reconfigured to become educational platforms that foster learning and sensemaking [11, 13–15].

1.1 Knowledge and Learning

Educational and Cognitive scientists have discovered that learners retain knowledge better, and are able to apply it to a wide range of contexts, when meaningful learning takes place. Ambrose and colleagues [1] have proposed several principles of meaningful learning that are “domain independent, experience independent, and cross-culturally relevant”. Some of these principles are as follows: (i) students' **prior knowledge** can help or hinder learning. (ii) how students **organize knowledge** influences how they learn and apply what they know. (iii) students' **motivation** determines, directs, and sustains what they do to learn. (iv) to become **self-directed** learners, students must learn to **monitor and adjust** their approaches to learning. In other words, **motivation**, **metacognition**, and **self-regulation** are important individual differences that moderate and foster meaningful learning.

The Internet, digital media technologies, and e-learning ‘ecologies’ enable new forms and models of learning, that are fundamentally different from the traditional classroom practices of didactic pedagogy [10]. Learners are no longer passive *consumers* of knowledge, but also *modifiers*, *producers*, and *exchangers* of knowledge. In **active knowledge making** [2], learners can, and often need to, find information on their own using online resources. They are not restricted to the textbook alone. The Internet is often a definitive resource for information on any given topic [55]. A learner can search the web (to learn) at any time, from anywhere, on any web-enabled device. As knowledge producers, learners search and analyze multiple sources with differing and contradictory perspectives, and develop their own observations and conclusions [12]. This is a skill that must be learned, as many sources of information

	ENTRY SURVEY [SUR1]	INITIAL SESSION [SES1]	LONGITUDINAL TRACKING [SES2a, SES2b, SES2c, SES2d]	MID-TERM SURVEY [SUR2]	FINAL SESSION [SES3]	EXIT SURVEY [SUR3]
Why	Record individual-differences	Establish baseline search behaviour and initial knowledge	Understand change in search behaviour and knowledge acquisition over time	Track changes in individual differences	Record "evolved" search behaviour, and "final" knowledge	Final state of individual differences
When	Week 1-2 of semester	Weeks 1-2 of semester; after SUR1	Four different points over the semester	Semester mid-point	After last day of classes	Anytime after SES3
Where	Asynchronous	Synchronous: Remote	Async	Async	Sync: Remote	Async
What	<u>Only in SUR1:</u> -Consent Form -Search Exp. & IT proficiency <u>Repeated in SUR2 and SUR3:</u> -Course Load -Note-taking strategies -Motivation -Self-regulation -Metacognition	Two search tasks: for each task, participants searched to find at least three unique, good quality online resources relevant to a given topic. • Pre-search self reporting: existing knowledge, interest, perceived difficulty • Post-search self reporting: perceived learning, perceived search success, interest and motivation, decision making One website reliability assessment from Stanford History Education Group (SHEG)	Participants <u>recorded</u> browsing activity when they worked on final project assignment – writing a research paper – at four different points in the semester. - SESa: Proposal - SES2b: Paper Outline - SES2c: Rough Draft - SES2d: Final Paper Participants also shared (anonymized) assignment submission	Similar to SUR1, with repeated components	Two search tasks: one task-topic repeated from SES1, one new; same format as SES1 One website reliability assessment from SHEG (topic different from SES1) Semi-structured interview: reflection on searching and learning experience.	Similar to SUR2 Participants self-reported scores and grades they received for different parts of the final project
Approx. Time Req.	10 - 15 mins	60 - 90 mins	No time limit for working on assignments. Sharing data with researchers took 1-5 minutes.	10 - 15 mins	60 - 90 mins	10 - 15 mins
Comp: (USD) \$150	\$5	\$25	\$5, \$5, \$10, \$15 (total \$35)	\$10	\$30	\$15
Bonus \$30 paid in the end, if participant completed all parts of the study.						

Figure 1: Exploratory longitudinal study procedure, proposed in Fall 2021, and conducted in Spring 2022 semester, at the School of Information, University of Texas at Austin.

are not valid, reliable, or authentic [30, 53]. Understanding the different sources and identifying the more reliable ones are essential for effective teaching and learning [29, 31]. This is a critical aspect because the inability to cite properly or to use reliable resources provides learners with misconstrued knowledge [2, 6, 31].

1.2 Information Searching

Information search behaviour [52] can be broken down into three components: (i) **query formulation**, or generating and submitting search queries to a system (ii) **source selection** or examining a list of returned Search Results, and (iii) **interacting with sources**, or evaluating individual content pages [28, 41, 44]. Prior work on domain knowledge and expertise have linked domain expertise and search behaviour in terms of metrics, behavioural patterns, and criteria [8, 27, 34, 48, 50].

In general, with increased knowledge, searchers use more domain-specific terms in their queries, spend less time to assess relevance of search results, and focus more on the content pages. Wildemuth [51] showed that novices converge toward the same search patterns as experts, as they are exposed to a topic and learn more about it.

Zhang, Cole, and Belkin [54] found that features such as document retention, query length, and the average rank of results selected could be predictive of domain expertise. Cole and colleagues [8] further found that eye-gaze patterns could be used to predict an individual's level of domain expertise using estimates of cognitive effort associated with reading. White and colleagues [50] showed that measures such as diverse website visitation, more narrow topical focus, less diversity (or entropy), more 'branchiness' of search sessions, less dwell time, and higher query and session complexity are indicative of expert knowledge and/or search behaviour. A representative summary of these behaviours is present in [4, Figure 3], which has been adapted from literature reviews by [36] and [44].

Higher education researchers have also been increasingly studying how students' information search and information use behaviour affect and support their learning [46, 47, 55]. Literature on higher education students' information search behaviour was reviewed by Zlatkin-Troitschanskaia et al. [55]. Students can be considered as novices in three respects: domain/subject-matter, search skills, and task. The authors report that across literature, higher education students' information search behaviour tends to

follow some general patterns: (i) *foraging*: no explicit (task-specific) research plan and little understanding of the differences (pros/cons) between various IR systems; (ii) *Google dependence*: no intention to use any search tool other than Google; (iii) *rudimentary search heuristic*: reliance on one and the same simple search strategy, regardless of search context; (iv) *habitual topic changing*: students change the search topic after rather superficial skimming, and before evaluating all search results; and (v) *overuse of natural language*: students type questions into the search box that are phrased as if posing them to a person. These growing interests in the intersection of searching and learning highlights the importance of understanding learning during search, in order to improve human-information interaction.

1.3 Problem Statement & Research Contribution

A major limitation of Information Seeking research, and parallelly in Human-Computer Interaction (HCI) research, is that the user is examined in the short-term, typically over a single lab session. Very few studies exist in the searching-as-learning domain that have observed the same participant over a longer period of time than a single lab session [18, 19, 24, 45, 50, 51]. Furthermore, most of these studies were conducted when search engines were not as ubiquitous as they are today. Higher education students are increasingly using the Internet as their main learning environment and source of information when studying. Yet, to the best of the author’s knowledge, almost no new longitudinal studies were reported in major searching-as-learning literature recently, that systematically studied students’ information search behaviour and information-use over the long term, in their in-situ naturalistic environments and contexts, and linked those behaviours to the students’ learning outcomes and individual differences. Researchers across disciplines have expressed severe concerns over the short term nature of today’s searching-as-learning, and more broadly, HCI research [17, 20, 22, 55]. This ephemeral approach has acute implications in any domain where learning is involved, because “learning is a process that leads to change in knowledge ... (which) unfolds over time” [1], and “... does not happen all at once” [49].

To address the gaps in our knowledge on how information searching behaviour influences students’ learning processes over time, we conducted a semester-long exploratory longitudinal study with university student participants. The overarching research aim are to identify how students’ information searching behaviour correlate with their learning outcomes for a particular university course. Building upon principles from the Learning Sciences [1, 32, 33, 39], and empirical evidences from the Information Sciences [36, 44, 48], our study aims to (i) situate students as learners in their naturalistic contexts, and characterized by their individual differences (motivation, metacognition, self-regulation, technical skills, etc.); (ii) measure students’ information search and information use behaviour over time, and (iii) correlate the information search behaviour with the learning outcomes for the university course.

Conducting a longitudinal study is a very non-trivial endeavour, requiring meticulous planning and execution. Therefore, we sought the guidance and mentorship from the research community at various stages of the study planning – a doctoral consortium [3], and a dissertation proposal [4]. While these works were in the

planning phase of the study, our main contribution in this paper are the dissemination of preliminary findings: latent profile analyses to identify groups of participants, and investigating differences in information search behaviour between these groups. We also propose the name “LongSAL” to refer to this study for the first time.

2 RESEARCH METHODOLOGY

2.1 LongSAL: the Longitudinal Study

LongSAL (Longitudinal Search as Learning) is a remote, exploratory, longitudinal study conducted between January and June 2022 (Spring semester) at the School of Information, University of Texas at Austin. Participants were recruited from the student pool enrolled in the required undergraduate core-course: *Ethical Foundations for Informatics* [12]. 18 participants originally signed up for the study; 10 participants fully completed all the stages of the study, and the remaining 8 dropped off at different points during the semester. Students enrolled in the course had to submit a research paper of 2,000-2,500 words as the final project for the course. There were four checkpoints spread across the semester to submit the drafts in progress: (i) paper proposal, (ii) outline, (iii) rough draft, and (iv) final paper. Writing the research paper required choosing an informatics ethical dilemma, and applying three ethical perspectives covered in the course to explore potential solutions to the selected dilemma. This involved searching for information online, combining ideas, and citing at least 20 external sources.

The longitudinal study consisted of six data collection stages (Fig. 1). They comprise three asynchronous **surveys** (SUR1, SUR2, SUR3), two remote **synchronous sessions** over Zoom video conferencing software (SES1, SES3), and a set of four asynchronous **longitudinal tracking sessions** (SES2a, SES2b, SES2c, SES2d). The three asynchronous surveys measured individual differences, or moderating variables of the participants, in the beginning (SUR1), middle (SUR2), and end of the semester (SUR3). The components of the surveys are adapted from established instruments, viz. **Intrinsic Motivation Inventory** (IMI) [37], **Self-Regulation Questionnaire** (SRQ) [7], and **Metacognitive Awareness Inventory** (MAI) [40, 42, 43]. These survey components help to understand the students’ contexts, and situate them as learners performing long-term information searching (Sec. 1.1). The two synchronous sessions, SES and SES3, were online observations with screen, audio, and search-log recording. Participants performed two search tasks, and a website reliability assessment from the Stanford History Education Group (SHEG) [6]. The four asynchronous longitudinal tracking sessions, SES2a, SES2b, SES2c, and SES2d, were remote search log data collections, performed by participants whenever they worked on the final project research paper.

The primary tools for data collection were Qualtrics for surveys, and YASBIL for interaction log data. YASBIL (Yet Another Search Behaviour and Interaction Logger) is a two-component logging solution for ethically recording a user’s browsing activity for Interactive IR user studies [5]. YASBIL comprises a Firefox browser extension and a WordPress plugin. The browser extension logs the browsing activity in the participants’ machines. The WordPress plugin collects the logged data into the researcher’s data server. The logging works on any webpage, without the need to own or have knowledge about the HTML structure of the webpage. YASBIL

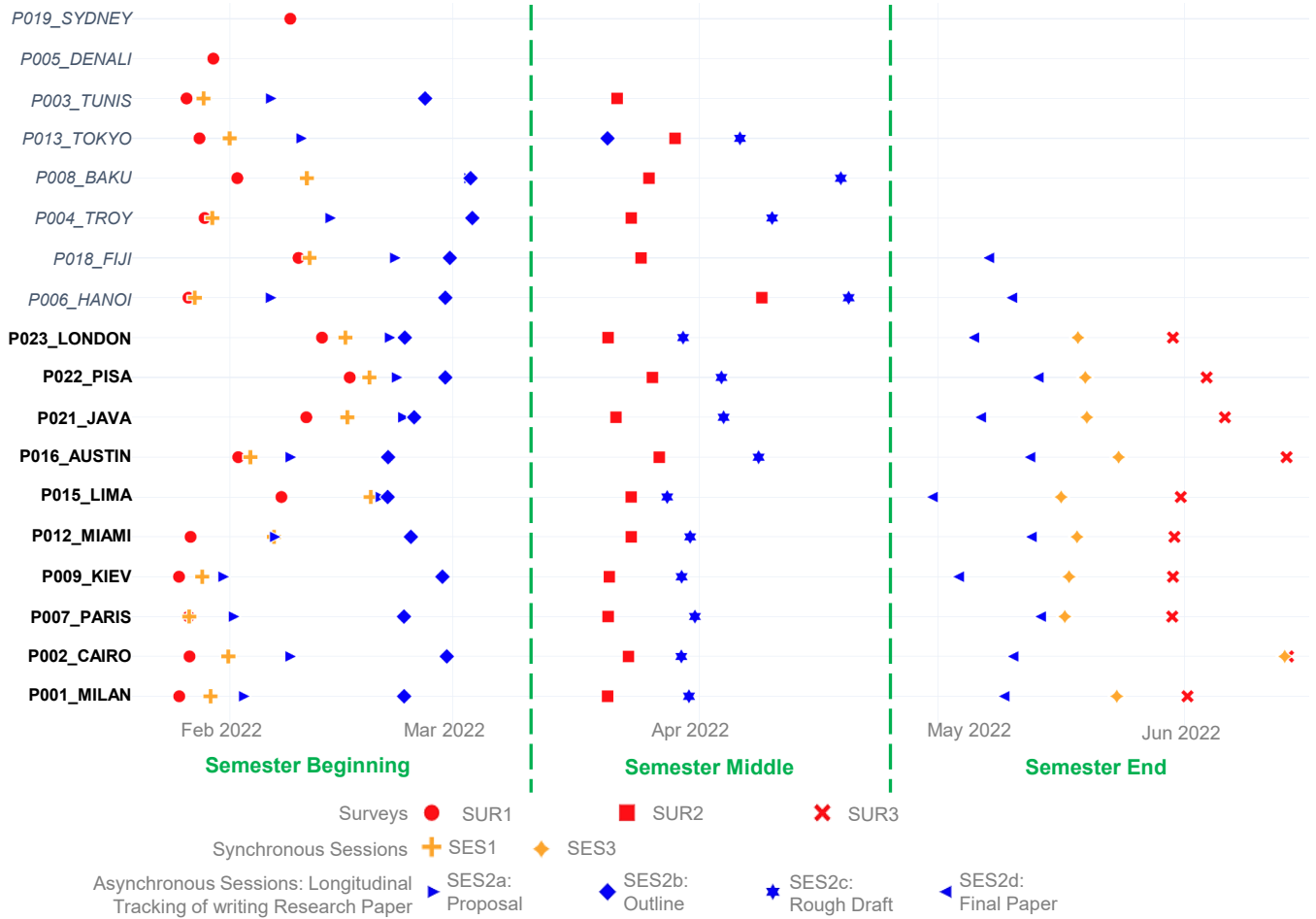


Figure 2: Timeline of collected data spread across the Spring 2022 semester. Participant names in bold (bottom 10) fully completed the longitudinal study, while participant names in *italics* (top 8) dropped off at various stages along the way. For data analysis, we have partitioned the semester duration into three phases (green vertical dotted lines): beginning, middle, and end of the semester. Inspired by the popular TV series, *La Casa de Papel* (English: “Money Heist”), participants were assigned geographic code-names.

also offers ethical data transparency and security towards participants, by enabling them to view and obtain copies of the logged data, as well as securely transfer the data to the researcher’s server. Participants received repeated reminders to turn YASBIL on only when they were searching for information related to the course.

3 PRELIMINARY RESULTS AND DISCUSSION

3.1 Latent Profile Analysis

As discussed in Sec. 1.1, motivation, metacognition, and self-regulation are important factors that affect students’ searching as learning process. We are interested in understanding subgroups of participants who share commonalities in these factors. These subgroups often cannot be captured directly but instead must be inferred from other information [16]. Latent profile analysis (LPA) is a statistical technique used to identify underlying subgroups, or “latent profiles” within a population, based on a set of observed variables

[9, 35, 38, 56]. LPA provides a framework to describe population heterogeneity in terms of differences across individuals on a set of characteristics, as opposed to describing the variability of a single variable. This makes LPA a **person-centred approach** [25], as opposed to traditional variable-centred approaches such as multiple regression analysis or factor analysis. The underlying principle of person-centered approaches is that, rather than quantifying the role of particular variables in a study, a population is organized into a finite number of mutually exclusive and exhaustive subgroups, each comprising individuals with similar characteristics. In other words, each latent profile represents a subgroup of individuals characterized by a pattern of responses on a set of variables. Therefore, LPA can be used to identify and describe the optimal number of latent profiles to represent a population. Latent Profile Transition Analysis (LPTA) is the extension of LPA to longitudinal designs (as in our case), which additionally identifies whether participants change

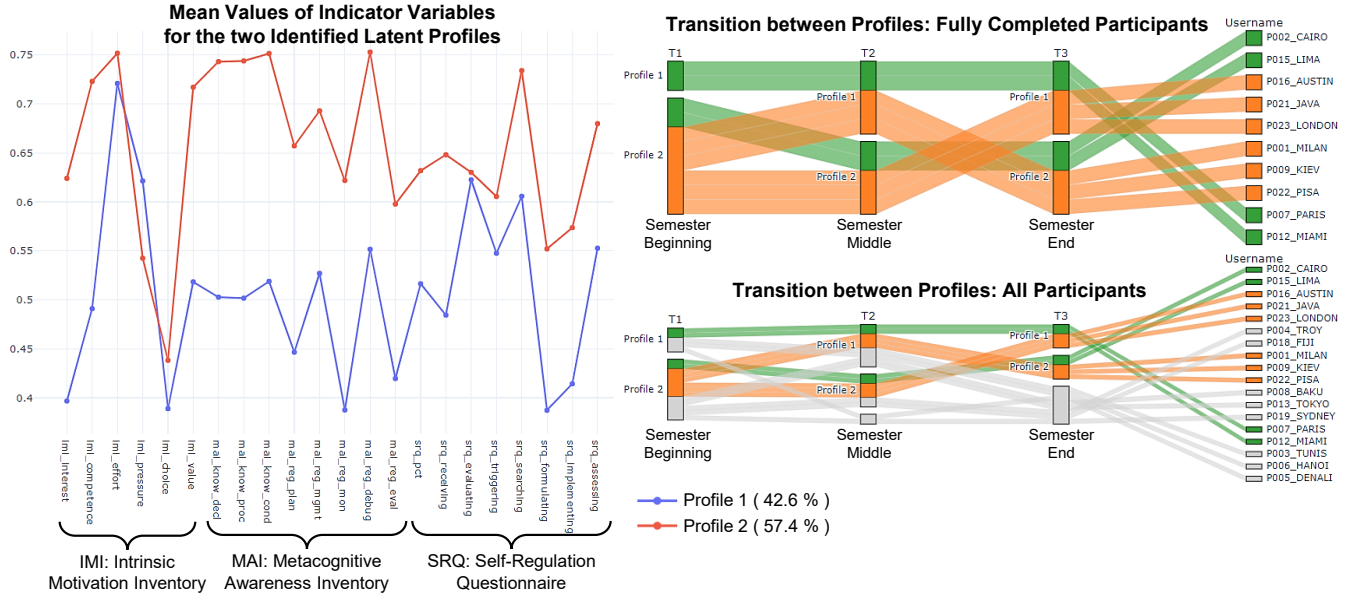


Figure 3: Latent Profile Analysis identified two profiles of participant characteristics, based on the self-reported values of motivation (IMI), metacognition (MAI), and self-regulation (SRQ).

profile membership (i.e., transition) between different time-points during data collection.

We captured motivation, metacognition, and self-regulation at three different points in the semester (Fig. 2): beginning (SUR1), middle (SUR2), and end (SUR3). Therefore, we conducted Latent Profile Analysis, and then transition analysis of profile memberships at these three different timepoints. Two latent profiles were identified. The results are shown in Fig. 3. The line chart on the left of Fig. 3 shows the normalized mean values of the identified profiles. Profile 1 is characterised by generally lower values of motivation, metacognition, and self-regulation, whereas Profile 2 is characterised by higher values. The differences are more prominent for the metacognition and self-regulation components, than for the motivation component. The transition diagrams on the right in Fig. 3 show whether participants stayed in the same profile throughout the semester (green bands), or whether they transitioned to a different profile (orange bands). Grey bands indicate participants who did not complete the entire study

3.2 Webpage Visit Count and Dwell Time

After identifying the two latent profiles (Sec. 3.1), we investigated how participants belonging to different profiles differed in their information search behaviour. In this preliminary analysis, we report on four measures of webpage visits: (i) count of unique webpages visited, (ii) total dwell time (iii) average dwell time, and (iv) variability (standard deviation) of dwell time, on different categories of webpages.

Information search behaviour can be broken down into three components: query reformulation, source selection, and interacting with sources (Sec. 1.2). Accordingly, we decided to test differences in visit count and dwell time on three categories of webpages: (i) those that have a query input interface (e.g., Google homepage,

search page of a library or academic database, such as ACM Digital Library); (ii) webpages presenting a list of information objects to the participant – such as search engine results page or academic database results page; and (iii) content pages, which contain significantly more information than a list of links. These webpage categories were identified from URL patterns, the open-graph meta tags¹ in the source HTML of the webpage, and by manual inspection. Since the measures were not normally distributed, we employed non-parametric Mann-Whitney U tests to examine differences between the profiles, reflected in these measures. The results are reported in Table 1, and significant differences are marked.

The results indicate that participants belonging to Profile 1, characterized by lower levels of metacognition and self-regulation, demonstrated a tendency to visit more webpages across all categories, in comparison to participants in Profile 2, characterized by higher levels of metacognition and self-regulation. Notably, participants in Profile 1 consistently spent a significantly greater amount of time on webpages, both in total and on average, and exhibited a higher degree of variability in their dwell time. These findings suggest that participants in Profile 2 demonstrated greater efficiency in their searching-as-learning behaviour. The reduced variability in dwell time among this group may be indicative of a more consistent performance, while shorter dwell times may indicate a higher level of task focus, which is associated with higher levels of self-regulation [23]. Furthermore, visiting fewer webpages may indicate greater efficiency in the acquisition of knowledge, as participants in Profile 2 were able to gain the desired information from fewer content webpages, publications, and articles.

We also analysed the self-reported scores and grades received by the participants in these two profiles. There were no significant

¹<https://ogp.me/#types>

Table 1: Metrics of webpage visitations, with means (and standard deviations) of the two profiles identified using Latent Profile Analysis (Sec. 3.1). Bold numbers and green-shaded cells denote cases when the two groups differed significantly, at the 1% () significance level ($p < .01$), and 5% (*) significance level ($p < .05$). Differences approaching 5% significance ($.05 \leq p < .1$) are denoted by $\wedge$ and yellow-shaded cells.**

Webpage Type	Count of Unique Webpages Visited		Total Dwell Time (seconds)		Average Dwell Time (seconds)		Variability (SD) of Dwell Time (seconds)	
	Profile 1	Profile 2	Profile 1	Profile 2	Profile 1	Profile 2	Profile 1	Profile 2
Query Formulation								
Overall	1.5 (0.5)	1.1 (0.3)	11.1 (5.8)	7.5 (1.5)	9.5 (3.7)	7.5 (1.5)	0.5 (0.5)	0.4 (1.6) $\wedge$
Webpages for Source Selection (List of Sources)								
Overall	2.8 (2.2)	1.8 (0.8) $\wedge$	174.9 (138.9)	81.7 (61.0) **	70.5 (67.2)	41.8 (17.4) *	29.0 (32.1)	13.5 (16.2) *
Web SERPs	1.9 (1.7)	1.2 (0.6) $\wedge$	71.7 (29.2)	55.8 (35.8) *	56.7 (26.5)	41.3 (10.5) *	3.9 (4.8)	2.2 (5.5) $\wedge$
Publication Search	3.4 (2.7)	1.0 (0.0)	191.8 (174.8)	16.5 (0.5) *	52.3 (34.6)	32.9 (38.1) *	47.4 (42.0)	17.9 (36.5) **
Content Webpages (Interacting with Sources)								
Overall	7.8 (7.6)	4.5 (3.7) $\wedge$	345.4 (172.5)	215.6 (105.5) $\wedge$	250.2 (128.2)	177.4 (75.1) $\wedge$	41.4 (59.2)	14.5 (26.7) *
Webpages (Generic)	2.0 (1.2)	1.2 (0.4)	75.0 (12.6)	105.2 (28.3)	27.2 (23.5)	34.4 (16.6) $\wedge$	2.4 (5.3)	6.3 (7.3)
Publications	2.0 (1.0)	1.4 (1.0) $\wedge$	224.9 (111.5)	130.6 (107.0) *	119.6 (59.6)	88.2 (21.3)	9.6 (9.9)	19.3 (15.5) *
Books	1.2 (0.4)	1.5 (0.5)	16.0 (0.0)	22.0 (2.0)	13.5 (2.3)	24.0 (0.0)	0.0 (0.0)	0.9 (1.8)
Articles (as per HTML meta tag)	3.1 (2.4)	1.5 (1.0) **	129.4 (136.3)	76.6 (115.8) *	209.3 (62.3)	166.8 (80.6) *	17.3 (27.4)	3.5 (8.7) *

differences in the grades received. This further corroborates the efficiency hypothesis, that Profile 2 individuals were able to score the same grades despite visiting less number of webpages, and spending less time on webpages.

4 CONCLUSION AND FUTURE WORK

In this paper we discuss the very preliminary findings from the multitude of data collected in the LongSAL study. We conduct latent profile analysis on data about participants' individual differences, viz. motivation, metacognition, and self-regulation. Our results showed that participants having higher levels of metacognition and self-regulation were seemingly more efficient in their searching as learning behaviour: they visited less number of webpages, spent less time reading webpages, and were still able to get the same grade-scores as that of participants spending more time on webpages. Immediate next steps in our analyses will be the examination of search queries submitted by the participants, and perform query reformulation analysis [26]. We will also conduct qualitative analysis of the semi-structured interview conducted in SES3 (Fig. 1). We will also compare the differences in search behaviours between synchronous sessions : SES1 and SES3.

Our study has several limitations, the primary being the small sample size (10 fully completed participants, and 18 overall). The required core course had close to 100 enrolled students. Although we made significant efforts at participant recruitment, due to circumstances beyond our control, no more than 18 participants signed up for the study. Another major limitation is that our longitudinal tracking tool – YASBIL – works on webpages only. If participants were downloading and reading PDFs offline, we currently have no way to capture that learning behaviour.

The Internet and digital educational technologies offer great opportunities to transform learning and the education experience. Enabled by our increased understanding of longitudinal searching-as-learning processes, we can create a new wave of fundamentally transformative educational technologies which will be “more engaging for learners, ... more resource efficient, and more equitable in the face of learner diversity” [10].

ACKNOWLEDGMENTS

The LongSAL study would not have been possible without the willingness of our participants, especially for choosing to continue participating for a whole semester, despite the difficulties posed by the COVID-19 pandemic. Special thanks to the author's dissertation committee members, Professors: Jacek Gwizdka (chair), Soo Young Rieh, Matt Lease, and Robert Capra; and the author's mentors at the ACM CHIIR 2021 Doctoral Consortium, Professors: Pertti Vakkari, Ian Ruthven, and Gerd Berget, for their continuous guidance and mentorship. The work has been financially supported by research grants to Prof. Rieh and Prof. Gwizdka, the David Bruton Graduate Fellowship from the University of Texas at Austin, student travel grants from ACM SIGIR, and the 2022 ASIS&T Doctoral Dissertation Proposal Scholarship.

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